Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	26 October 2026
Team ID	Gyanvi Jain
Project Name	Formula 1 Race Intelligence Dashboard
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

Project Name: Formula 1 Race Intelligence Dashboard

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
FR-1	Data Access & Integration	The solution shall access and integrate Formula 1 race results, driver details, team information, circuit data, and championship standings from the available Excel/CSV datasets.	High
FR-2	Data Processing	The solution shall handle missing values, validate data types, join race results with circuit information, and calculate DNF counts and cumulative points.	High
FR-3	Business Analysis	The solution shall calculate key performance indicators, driver cumulative points, team championship performance, DNF statistics, tyre compound usage, and podium achievements.	High
FR-4	Business Insights & Analytical Outputs	The solution shall provide an interactive dashboard containing four KPI cards, a driver leaderboard, DNF reliability analysis, team performance charts, tyre usage analysis, a geographic participation map, and podium visualisations.	High
FR-5	User Interaction & Analysis	The solution shall allow users to filter data by DNF reason and race year, adjust the Top N parameter, and compare driver and team performance across available seasons.	High

FR-6	Data Update & Reporting	The solution shall allow the dashboard to be republished when the underlying dataset or analysis is updated and provide accessible visual summaries through Tableau Public, with optional embedding in a Flask web application.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard shall have clear chart titles, readable labels, understandable legends, and easy-to-use filters and parameters so users can explore Formula 1 statistics with minimal difficulty.
NFR-2	Security	The solution shall use publicly shareable, non-sensitive datasets and avoid displaying personal or confidential information. Access to any separately hosted application should be configured appropriately.
NFR-3	Reliability	The dashboard shall display consistent results based on the prepared dataset, handle missing values appropriately, and calculate KPIs and performance metrics accurately.
NFR-4	Performance	The dashboard should load visualisations and respond to filter selections efficiently when working with the available dataset of approximately 13,750 rows.
NFR-5	Availability	The published dashboard should be accessible through Tableau Public when the service and internet connection are available.
NFR-6	Scalability	The data source and dashboard structure should support adding more seasons, race results, drivers, teams, and circuits with appropriate data preparation and visualisation updates.